# Physics 1 - Problem Set 5: Energy

**Problem 2.**

A 2.0 kg block starts from rest at the top of a frictionless ramp that is 5.0 m high. It slides down the ramp and onto a level floor. What is the speed of the block at the bottom of the ramp? Use g = 9.8 m/s^2.

Answer: v = _____ m/s

**Problem 3.**

The same block then hits a spring (k = 400 N/m) on the floor. How far does the spring compress before the block stops?